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Activity - 2

1) CCTV Image with Salt and pepper Noise.

Technique used: Median Filter

Reason: It removes salt-and-pepper noise by replacing each pixel with the median of neighbori-ng pixels while preserving edges.

2) Satellite Image with Guassian Noise.

Technique Used: Gaussian Filtering

Reason: It reduces Guassian noise by smoothing pixel intensity variations and producing a cleaner image.

3) Medical X-ray with Random Noise and Fine Edges.

Techniques: Bilateral Filtering.

Reason: It removes noise while preserving important edges and fine medical structures

4) Scanned Document with Random Black and white Dots

Technique used: Median Filtering

Reason:- It remove unwanted black and white dots preserving text characters and document details.

5) Low-Light Image with heavy Grain

Technique used:- Non-Local Means Denoising

Reason:- It reduces heavy grain and random noise while preserving useful image details & structures.

6) Foggy Road Image with very Low Contrast.

Technique used:- Histogram Equalization

Reason:- It improves image contrast by redistributing intensity values, making objects more visible.

7) Underwater Image Appearing Dull and Washed Out

Technique:- White Balance and color Correction

Reason:- They correct color distortion, restore natural colors, improve the overall appearance of the underwater image.

8) chest X-Ray with poor visibility of internal structures

Technique : Contrast Limited Adaptive Histogram Equalization (CLAHE)

Reason :- It enhances local contrast, making internal structures more visible without excessively amplifying noise.

9) Night-Time Traffic Image Lacking intensity Variation.

Technique :- Gamma Correction

Reason :- It brightens dark regions and improves the visibility of vehicles and other objects without excessively brightening entire image.

10) Grayscale Image with Narrow Intensity Range.

Technique :- Contrast Stretching

Reason :- It expands the narrow intensity range across a wider range of gray levels, improving image contrast and visibility.